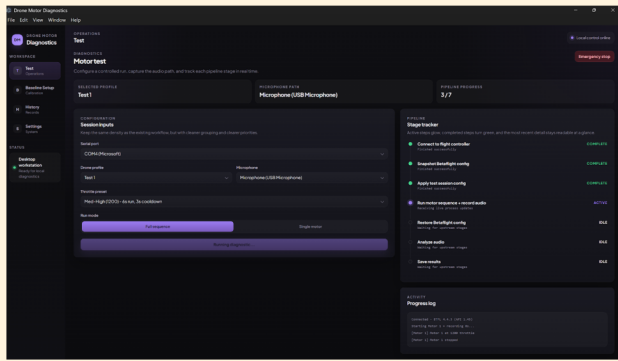




Key Design Decisions

Decision 1 — Brand new motor as healthy baseline

- A brand new motor was purchased as a healthy reference
- Used instead of original motor due to uncertainty of the old motor's health
- Generated a healthy FFT baseline to compare against 4 other drone motors

Decision 2 — Enclosure material selection

- Outer shell → half-inch plywood to block noise
- Inner lining → SONOPanX 7/16" fibreboard to absorb sound
- Fibreboard → reduces internal reflections to prevent FFT distortion
- Plywood → main noise isolation barrier

Next Steps

- Use thicker enclosure materials to improve sound dampening and reduce external noise interference
- Create airtight seal on the box when it is closed. Can be achieved through a gasket sealer or any rubber type seal
- Upgrade to a higher-sensitivity microphone to detect subtler frequency deviations
- Replace the single-motor healthy baseline with a large dataset of healthy motor signatures
- Collect data across multiple units, operating conditions, and wear states or find more data externally
- Train a machine learning model on this dataset to better distinguish faults from normal variation
- Transition from a hand-tuned $k \times \sigma$ threshold approach to a data-driven classification model

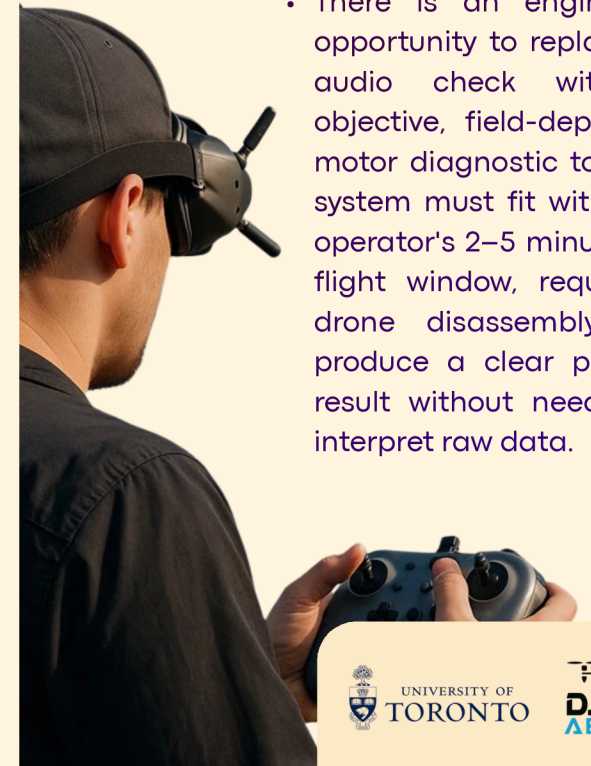
MOTOR FAULT DETECTION

via sound analysis



The Opportunity

- Professional FPV Drone cinematography drones' motor health is currently checked in a very subjective way. You hold the drone to your ear, if it sounds smooth, it flies.
- There is an engineering opportunity to replace the audio check with an objective, field-deployable motor diagnostic tool. The system must fit within the operator's 2–5 minute pre-flight window, require no drone disassembly, and produce a clear pass/fail result without needing to interpret raw data.



The Concept

FPV Drone

- Turn on one motor on at a time
- Connected through beta flight configurator
- GEPRC GRI404 4500KV Motor
- Diameter: 14 mm, weight: 9.8 g

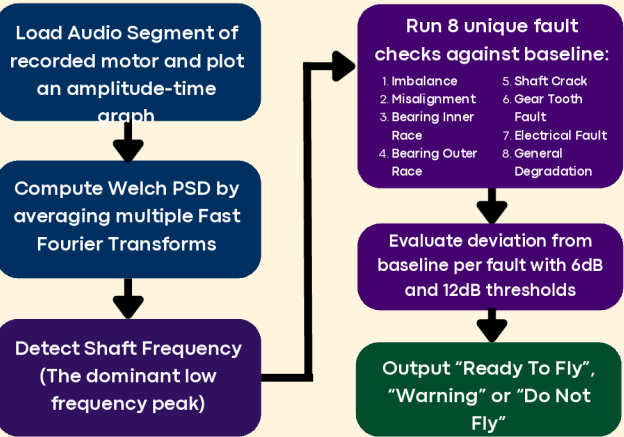
Sound Proof Box

- Made of half-inch plywood materials
- SONOPan X design absorbing impact sound
- 8 inch by 12 inch by 12 inch exterior

Software

- Betaflight Configurator

Software Architecture



Why Sound Analysis?

Eliminated Designs:

Vibration Analysis (FC Blackbox)

Every test requires launching Betaflight, connecting via USB, and completing an actual test flight — making it too time consuming to use as a pre-flight check. More critically, gyroscope readings vary significantly between motors and mounting positions, making it extremely difficult to establish a consistent healthy baseline.

Current Signature Analysis (MCSA)

Accessing the phase current requires tapping into the motor terminals, which are soldered directly to the internal flight controller board — a modification that risks permanent damage to the drone. Even if access is achieved, minor fluctuations in battery voltage directly affect current readings, meaning results shift between tests and a stable, repeatable baseline cannot be guaranteed in field conditions.

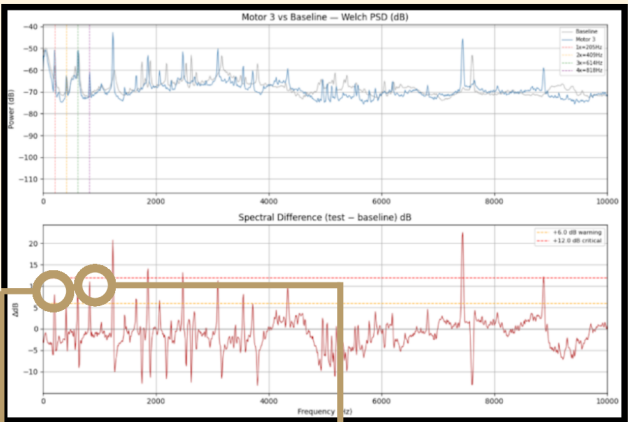
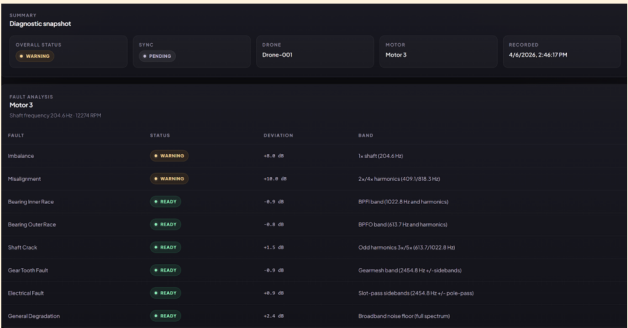
Verification

- **Signal acquisition** — Recorded audio from each motor individually in the enclosure and applied FFT analysis → successfully extracted measurable features (frequency, harmonics, peak count, band energy)
- **Fault detection** — Compared each motor's spectral features against the healthy baseline → Motor 3 flagged for misalignment/looseness (see next side for graph)
- **Baseline calibration** — Established a healthy reference using Motor 1 and compared all other motors against it → deviations in frequency content and harmonics aligned with expected fault signatures
- **Sensitivity to faults** — Analyzed changes in FFT outputs across motors → detected increases in peak count (6 → 9), higher harmonic ratios, and energy shifts to higher frequency bands in faulty motors
- **Response time** — Timed audio analysis algorithm → achieved ~3–4 seconds, well below 15 s requirement
- **Non-invasive installation** — Verified setup by running tests with no physical modification (no soldering/drilling) → system operates using external microphone only
- **Network independence** — Ran full diagnostic cycle in airplane mode → confirmed zero external network calls
- **Output clarity** — Presented results to users as simplified classifications (Ready to Fly / Warning / Do Not Fly) → no interpretation of raw data required
- **Repeatability** — Performed multiple runs under identical conditions → consistent feature extraction observed

Research-Backed Approach

All our results are based on Secondary Research. Our primary source is "Vibration-based Condition Monitoring: Industrial, Aerospace and Automotive Applications by Robert Bond Randall, published by John Wiley & Sons, 2011," which gives insight into how to read the results in the Welch PSD Graphs and associate observations with specific faults.

See the results for Motor 3 below, which has Warnings for Imbalance & Misalignment



Imbalance

On the spectral difference plot we can see Motor 3's 1x shaft peak is significantly higher than the baseline peak. This passes the orange line. This is directly from Randall, "...the responses will usually be distorted from sinusoidal and thus contain some level of harmonics of the fundamental frequency."

Misalignment

On the spectral difference plot we can see Motor 3's shaft peak is significantly higher than the baseline peak at the 2x and 4x harmonics of the shaft frequency. This comes from Randall, "the second and fourth harmonics were quite high...the increases in vibration were due to increasing misalignment."